# Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

| Role | Name | Email |
|---|---|---|
| **Developer (You)** | Nathaniel Sutton | Naters@udel.edu |
| **Friend (optional)** | | |

## Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

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| *I want to create a website related to my major which is civil engineering. I want to make a calculator type website that helps people calculate how much concrete they need for a project and how much of each ingredient they would need if they were to mix their own concrete.* |

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

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| *An example of a current website that also can measure the amount of concrete needed for a project is a page on https://www.measureandbuild.com/concrete-mix-calculator/./* |

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

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| *By entering the length width and depth of a desired concrete slab or formation, the website will calculate how much concrete is needed. It will also calculate how much of each ingredient is needed to mix into the concrete. I would want this website to be used by people* |

*working on small projects that need help measuring concrete size.*

# Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal is to take the dimensions of a concrete slab, calculate the volume and the ingredients needed to fill the slab.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will input the length and width in feet and the depth in inches of the slab.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It will convert the depth to inches so that it could calculate the volume of the slab. It will use a pre set mix ratio to determine the amount of each ingredient needed.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

It will display the total volume of concrete needed in cubic feet and the amount of each ingredient needed.

# Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

*The program needs to track concrete projects, measurements, and materials. Project represents the slab being calculated, measurements represent dimensions, and materials represent the ingredients needed.*

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

*Length: Length of slab in feet*
*Width: Width of slab in feet*
*Thickness: Thickness of slab in inches*
*Volume: Calculated volume of concrete needed.*

# Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

*Some parts of this project that seem the most difficult or challenging are Drafter and measurements. I would need to use dataclasses and lists and put them through drafter. The measurements could get confusing especially when converting inches to feet.*

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

*I will be using an LLM to help me code this project. I will ensure that i understand the generated code by asking questions and fully reading through and understanding the code before trying to utilize it.*

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

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| *My ultimate goal for this project is to create a website that combines computer science and my own major of civil engineering.* |

# Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

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| *I want to create a website using python and drafter. The website should combine computer science concepts with civil engineering concepts. I want the website to take the dimensions of a concrete slab and calculate how much concrete is needed to fill it and how much of each ingredient within the concrete is needed.* |

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

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| *It told me that i would need alot more dimensions in order to measure for things that aren't slabs and that i should import certain formulas to help with the mix ratio.* |

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

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| I need the project to be simple, things that aren't slabs will be too complex in measuring. Lets stick with only rectangular slabs. |

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

*My project idea has been more focused on simpler designs. But overall it hasn't changed much.*

# Part 6: Resource Planning

1. What resources will you use if you get stuck?

*If i get stuck during this project, i will utilize office hours, LLMs, and discord.*

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

1. *Read the error code*
2. *Check for misspelling*
3. *Use unit tests*
4. *make small changes*
5. *Run code again*

# Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

*I did not use a LLM to assist me with this assignment.*

# Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.